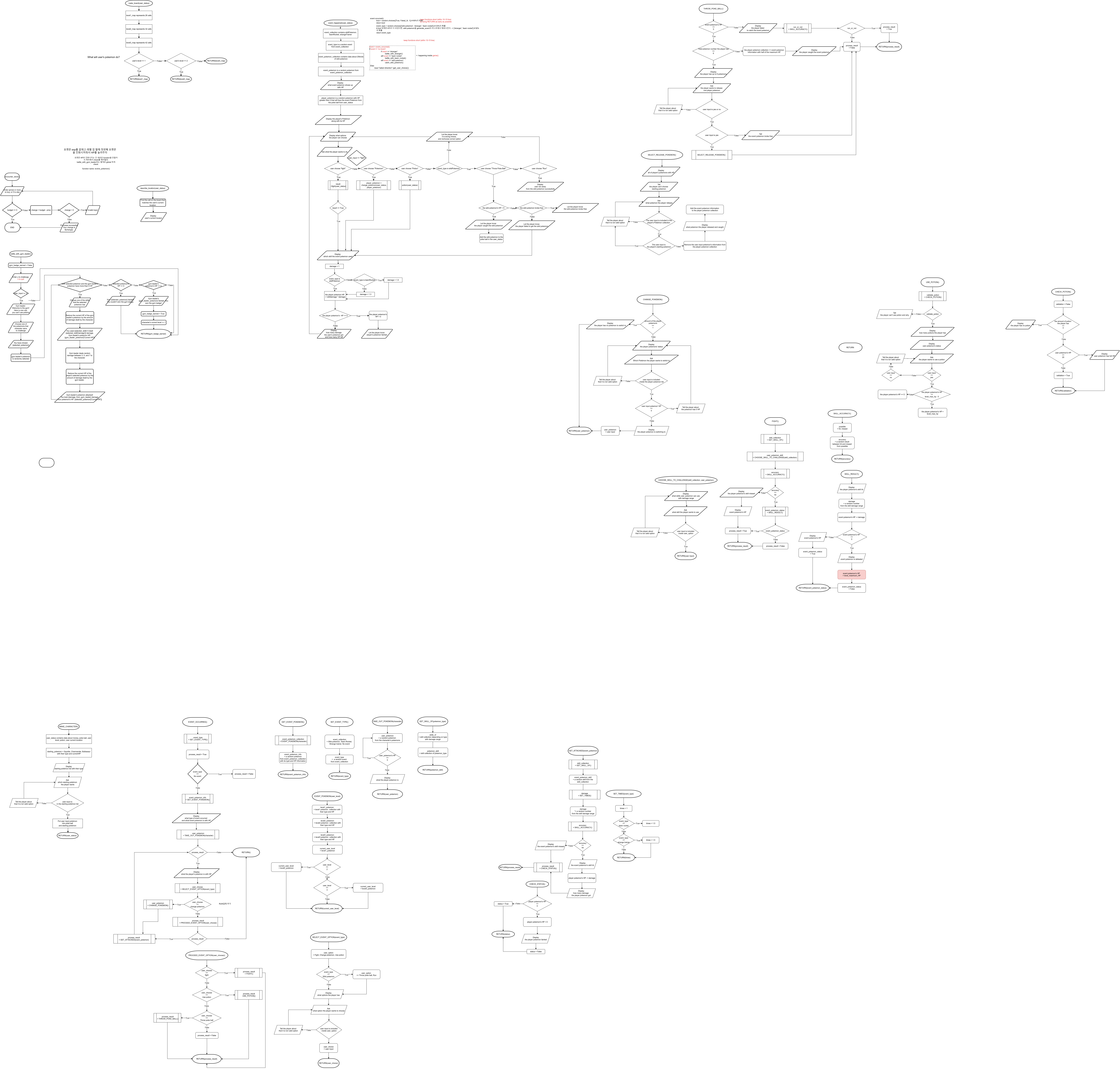
make_board(user_status)
level1_map represents 26 cells
level2_map represents 32 cells
level3_map represents 42 cells
What will user's pokemon do?
user's level == 1
False
user's level == 2
False
RETURN(level3_map)
True
RETURN(level1_map)
True
RETURN(level2_map)
encounter_store()
Enter {price} or more to buy, or 0 to skip
budget == 0
False
change = budget - price
change < 0
True
Invalid input.
True
False
END
Purchase successful! Your change is ${change}.
describe_location(user_status)
Find the cell on the board that matches the user's current location
Display user's current locaion
battle_with_gym_leader()
gym_badge_earned = False
Enter y to challenge, n to quit
user_input == 'y'
True
False
Gym leader: Welcome to the gym! Here is one rule, you can't use potions
Choose one of the pokemons that character owns to challenge
You have chosen {selected_pokemon}
gym leader's pokemon is randomly selected
both selected pokemon and the gym leader's pokemon have more than 0 HP
False
selected pokemon's HP <= 0
False
gym leader's pokemon's HP <= 0
False
True
Choose one of the skills that the selected pokemon has
Reduce the current HP of the gym leader's pokemon by the amount of damage dealt by the character
You used {selected_skill}! It dealt {selected_skill['damage']} damage. Gym leader's pokemon HP: {gym_leader_pokemon['Current HP']}
Gym leader deals random damage between 3, 5, and 7 to the character
Reduce the current HP of the player's selected pokemon by the amount of damage dealt by the gym leader.
Gym leader's pokemon attacked! You took {damage_from_gym_leader} damage. Your pokemon's HP: {selected_pokemon['Current HP']}
True
Your {selected_pokemon} fainted. You couldn't won the gym badge.
True
Gym leader's {gym_leader_pokemon} fainted. You won the gym badge!
gym_badge_earned = True
character's current level += 1
RETURN(gym_badge_earned)
event_happened(user_status)
event_collection contains wildPokemon, teamRocket, strangeTrainer
event_type is a random event from event_collection
event_pokemon_collection contains data about 20kinds of wild pokemon
event_pokemon is a random pokemon from event_pokemon_collection
Display what event pokemon shows up with HP
player_pokemon is a random pokemon with HP greater than 0 that will face the event Pokémon from the poke ball from user_status
Display the player's Pokémon along with its HP
Display what options the player can choose
Ask what the player wants to do
user_input == "fight"
user choose "fight"
False
user choose "Pokemon"
False
user choose "Potion"
False
event_type is wildPokemon
True
user choose "Throw Poke Ball"
False
user choose "Run"
False
Let the player know it is wrong choice, and rechoose correct option
False
True
result =fight(user_status)
True
player_pokemon = change_pokemon(user_status, player_pokemon)
True
potion(user_status)
True
Display user ran away from the wild pokemon successfully
True
result == True
True
the wild pokemon's HP < 5
False
the wild pokemon broke free
True
Let the player know the wild pokemon broke free
True
Let the player know the player caught the wild pokemon
False
Let the player know the player failed to get the wild pokemon
Add the wild pokemon to the poke ball in the user_status
Display which skill the event pokemon used
damage = 1
event_type is wildPokemon
False
event_type is teamRocket
True
damage = 1.5
True
False
damage = 1.3
the player pokemon HP -= skilldamage * damage
the player pokemon's HP <= 0
True
the player pokemon's HP = 0
False
Display how many damage the user's pokemon got and how many HP left
Let the player know player's pokemon fainted
THROW_POKE_BALL()
event pokemon's HP <= 5
False
Display the player failed to catch the event pokemon
run_or_not = SKILL_ACCURACY()
run_or_not
True
process_result = True
True
False
Total pokemon number the player has == 6
False
the player pokemon collection += event pokemon information with half of the maximum HP
Display the player caught the event pokemon
process_result = False
RETURN(process_result)
True
Display the player has up to 6 pokemons
Ask the player wants to release one player pokemon
user input is yes or no
False
Tell the player about that it is not valid option
True
user input is yes
False
Tell the event pokemon broke free
True
SELECT_RELEASE_POKEMON()
SELECT_RELEASE_POKEMON()
Display all of player's pokemons with HP
Tell the player can't choose starting pokemon
Ask what pokemon the player release
The user input is included in the player's Pokémon collection
False
Tell the player about that it is not valid option
True
The user input is the player's starting pokemon
True
False
Remove the user input pokemon's information from the player pokemon collection
Display what pokemon the player released and caught
Add the event pokemon information to the player pokemon collection
CHANGE_POKEMON()
amount of the player pokemons == 1
True
Display the player has no pokemon to switch to
False
Display the player pokemons' status
Ask Which Pokémon the player wants to switch to
user input is included inside the player pokemon list
False
Tell the player about that it is not valid option
True
user input pokemon' HP == 0
True
Tell the player about the pokemon has 0 HP
False
Display the player pokemon is switching to
user_pokemon = user input
RETURN(user_pokemon)
USE_POTION()
validate_potion = CHECK_POTION()
validate_potion
False
Tell the player can't use potion and why
True
Display how many potions the player has
Display user pokemon's status
Ask the player wants to use a potion
Tell the player about that it is not valid option
False
user input == no
user input == yes
False
True
the player pokemon's HP >= level_max_hp - 5
False
the player pokemon's HP += 5
True
the player pokemon's HP = level_max_hp
RETURN
CHECK_POTION()
validation = False
the amount of potion the player has == 0
True
Display the player has no potion
False
user pokemon's HP == 20
True
Display user pokemon has full HP
False
validation = True
RETURN(validation)
FIGHT()
skill_collection = GET_SKILL_OF()
user_pokemon_skill = CHOOSE_SKILL_TO_CHALLENGE(skill_collection)
accuracy = SKILL_ACCURACY()
accuracy == hit
False
Display the player pokemon's skill missed
True
Display event pokemon's HP
event_pokemon_status = SKILL_RESULT()
process_result = True
True
event_pokemon_status
False
RETURN(process_result)
process_result = False
SKILL_ACCURACY()
possible = hit, missed
accuracy = a random result between hit and missed from possible
RETURN(accuracy)
SKILL_RESULT()
Display the player pokemon's skill hit
damage = a random number from the skill damage range
event pokemon's HP -= damage
event pokemon's HP <= 0
False
Display event pokemon's HP
event_pokemon_status = True
True
Display event pokemon is defeated
event pokemon's HP = level_maximum_HP
event_pokemon_status = False
RETURN(event_pokemon_status)
CHOOSE_SKILL_TO_CHALLENGE(skill_collection, user_pokemon)
Display what skills user_pokemon can use with damage range
Ask what skill the player wants to use
user input is included inside user_option
False
Tell the player about that it is not valid option
True
RETURN(user input)
MAKE_CHARACTER()
user_status contains data about money, poke ball, user level, potion, user current location
starting_pokemon = Squirtle, Charmander, Bulbasaur with their type and currentHP
Display starting pokémon list with their type
Ask which starting pokémon the player wants
user input is in the starting pokémon list
False
Tell the player about that it is not valid option
True
Put user input pokémon into poké ball and starting pokémon
RETURN(user_status)
EVENT_OCCURRED()
event_type = SET_EVENT_TYPE()
process_result = True
event_type == No event
True
process_result = False
False
event_pokemon_info = GET_EVENT_POKEMON()
Display what type of event occurred and what event pokemon is with HP
user_pokemon = TAKE_OUT_POKEMON(character)
process_result
False
RETURN()
True
Display what the player's pokemon is with HP
user_choose = SELECT_EVENT_OPTION(event_type)
user_choose == change pokemon
True
user_pokemon = CHANGE_POKEMON()
RUN일때 추가
False
process_result = PROCEED_EVENT_OPTION(user_choose)
process_result
True
process_result = GET_ATTACKED(event_pokemon)
False
PROCEED_EVENT_OPTION(user_choose))
user_choose == fight
True
process_result = FIGHT()
False
user_choose == Use potion
True
process_result USE_POTION()
False
user_choose == Throw poke ball
True
process_result = THROW_POKE_BALL()
False
process_result = False
RETURN(process_result)
GET_EVENT_POKEMON()
event_pokemon_collection = EVENT_POKEMON(character)
event_pokemon_info = a random pokemon from event_pokemon_collection with its type and HP information
RETURN(event_pokemon_info)
SET_EVENT_TYPE()
event_collection = Wild pokemon, Team Rocket, Strange trainer, No event
event_type = a random event from event_collection
RETURN(event_type)
TAKE_OUT_POKEMON(character)
user_pokemon = a random pokemon from the character's pokemons
user_pokemon's HP == 0
True
False
Display what the player pokemon is
RETURN(user_pokemon)
GET_SKILL_OF(pokemon_type)
skills_of = skill collection depending on type with damage range
pokemon_skill = skill collection of pokemon_type
RETURN(pokemon_skill)
EVENT_POKEMON(user_level)
level1_pokemon = level1 pokemon collection with their type and HP
level2_pokemon = level2 pokemon collection with their type and HP
level3_pokemon = level3 pokemon collection with their type and HP
current_user_level = level1_pokemon
user_level == 2
True
current_user_level = level2_pokemon
False
user_level == 3
True
current_user_level = level3_pokemon
False
RETURN(current_user_level)
SELECT_EVENT_OPTION(event_type)
user_option = Fight, Change pokemon, Use potion
event_type == Wild pokemon
True
user_option += Throw poke ball, Run
False
Display what options the player has
Ask what option the player wants to choose
user input is included inside user_option
False
Tell the player about that it is not valid option
user_choice = user input
RETURN(user_choice)
GET_ATTACKED(event_pokemon)
skill_collection = GET_SKILL_OF()
event_pokemon_skill = a random skill from the skill_collection
damage = SET_TIMES()
damage *= a random number from the skill damage range
accuracy = SKILL_ACCURACY()
accuracy == hit
False
Display the event pokemon's skill missed
True
Display the event pokemon's skill hit
player pokemon's HP -= damage
Display how many damage the player pokemon got
process_result = CHECK_STATUS()
RETURN(process_result)
CHECK_STATUS()
player pokemon's HP <= 0
False
status = True
True
player pokemon's HP = 0
Display the player pokemon fainted
status = False
RETURN(status)
SET_TIMES(event_type)
times = 1
event_type == team rocket
True
times = 1.5
False
event_type == strange trainer
True
times = 1.5
False
RETURN(times)

포켓몬 exp를 없애고 레벨 업 할때 첫번째 포켓몬을 진화시켜줘서 HP를 늘려주자

포켓몬 HP로 진화시키는 건 새로운 function을 만들어서 계속해서 check를 해야될듯.
battle_with_gym_leader와는 별개로 global 하게 check

function name: evolve_pokemon()

```
event occurred():
        bool = random.choices([True, False], [4, 1]) # 80%의 확률
        return bool
        event_type = random.choices(['wild pokemon', 'stranger', 'team rocket']) # 33%의 확률
        # (but 포켓몬 6마리 다 모았다면, wild pokemon을 generate_event의 리스트에서 제외시킨다. -> ['stranger', 'team rocket'] # 50%의 확률
        return event_type
```

keep functions short (within 10-15 line)
by doing RETURN as early as possible

keep functions short (within 10-15 line)

```
event = event_occurred()
if event != 'no event':
        if event == 'stranger':
                battle_with_stranger()
        elif event == 'team rocket':
                battle_with_team_rocket()
        elif event == 'wild pokemon':
                catch_wild_pokemon()
Else:
        input "select direction" (get_user_choice())
```

<- happening inside game()